

1) Not secure a distinguisher

On input s , generate two outputs x, y from $H(s)$.

Check if $x = y$.

If they are equal, then output 0 (indicating that the input s was likely 0 λ).

If they are not equal, then output 1 (indicating that the input s was likely non-zero).

The advantage of this distinguisher is non-negligible, because it can correctly distinguish between the two cases with a probability of $1/2$. Since H is not a secure PRG, it does not have the property that the output looks random for all possible inputs.

2) Not secure a distinguisher

A successful distinguisher can be constructed as follows:

Given an output of H , we can compute x and y by dividing the output into two equal parts. Then, we can use x and y as inputs to G and compare the outputs. If x and y are equal, then the output of H must have been generated from $H(0\lambda)$, otherwise, it must have been generated from $H(s)$ for some non-zero s .

The advantage of this distinguisher is $1/2$, because for a random non-zero s , the probability that x and y are equal is $1/2^{3\lambda}$.

3) Not secure a distinguisher

Choose a random seed s and generate the output of H , i.e. $x||y||z||w$.

Generate the output of G using the first part of the output of H , i.e. x .

Compare the second part of the output of H , i.e. $y||z||w$, with the output generated in step 2, i.e. $G(x)$.

If the output of H is a secure PRG, then the outputs from step 2 and step 3 should be indistinguishable. However, they are not, because they are different.

The advantage of this distinguisher can be calculated as the probability that the output of step 2 and step 3 are different, which is $1/2^{(3\lambda)}$.